View Reviews

Paper ID
982

Paper Title
Agentic AI for Plant Disease Diagnosis: An Autonomous Multi-Agent Framework Integrating Vision, Retrieval, and Reasoning

Track Name
Artificial Intelligence and Data Analytics

Reviewer #1

Questions

1. Technical content and scientific rigour (Rate the technical .content of paper)
4. High

2. 2. Novelty and Contributions
4. High

3. 3. Documentation (Rate the paper organize as per the IEEE format, clarity of text, figures and accurarcy of references)
4. High

4. 4. Comments to Authors
The manuscript presents a timely and well-motivated contribution by integrating Agentic AI with plant disease diagnosis through a multi-agent framework comprising vision, retrieval, reasoning, and orchestration modules. The proposed architecture effectively extends conventional image classification by incorporating explainability, uncertainty-aware decision making, and human-in-the-loop escalation, thereby improving the practical applicability of AI systems in precision agriculture. The experimental evaluation on both in-distribution and out-of-distribution datasets demonstrates the effectiveness of the proposed framework in reducing harmful automated decisions while maintaining high diagnostic performance.

To further strengthen the manuscript, the authors may consider citing the following recent work related to transformer-based plant disease classification:

Sounkhla, I., Dhir, R., Gupta, S., & Chouhan, S. A. (2026). ResNet-Transformer Hybrid Model for Classifying Apple Leaf Diseases. Procedia Computer Science, 283, 5329–5335.

Minor proofreading is recommended to improve language consistency and presentation quality. Overall, the paper is technically sound, well organized, and makes a meaningful contribution to trustworthy AI-enabled agricultural decision support. It is recommended for acceptance.

5. 5. Recommendation

2. Accept

8. Technical content and scientific rigour (Rate the technical content of paper)

4. High

9. Novelty and Contributions

4. High

10. Documentation (Rate the paper organize as per the IEEE format, clarity of text, figures and accurarcy of references)

4. High

11. Comments to Authors

The manuscript presents a timely and well-motivated contribution by integrating Agentic AI with plant disease diagnosis through a multi-agent framework comprising vision, retrieval, reasoning, and orchestration modules. The proposed architecture effectively extends conventional image classification by incorporating explainability, uncertainty-aware decision making, and human-in-the-loop escalation, thereby improving the practical applicability of AI systems in precision agriculture. The experimental evaluation on both in-distribution and out-of-distribution datasets demonstrates the effectiveness of the proposed framework in reducing harmful automated decisions while maintaining high diagnostic performance.

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12. Recommendation

2. Accept

Reviewer #3

Questions

1. Technical content and scientific rigour (Rate the technical .content of paper)

2. Average

2. 2. Novelty and Contributions

3. Above Average

3. 3. Documentation (Rate the paper organize as per the IEEE format, clarity of text, figures and accurarcy of references)

3. Above Average

4. 4. Comments to Authors

1. The evaluation of the agentic layer is relatively limited. While classification accuracy and proposed agentic metrics are reported, there is no statistical significance analysis or confidence intervals to demonstrate the robustness of the observed improvements.
2. The threshold values used by the Orchestrator (0.85, 0.55, 0.15, and 0.5) appear empirically selected, but the manuscript does not provide sensitivity analysis or justification showing how these values were chosen.
3. The OOD dataset contains only 190 images (5 images per class), which is relatively small for assessing generalization performance. A larger and more diverse field dataset would strengthen the conclusions.
4. Acronyms such as OOD, RAG, FedAvg, FedOpt, and FedMA should be consistently defined at first appearance and checked for formatting consistency.

5. 5. Recommendation

2. Accept

8. Technical content and scientific rigour (Rate the technical content of paper)

2. Average

9. Novelty and Contributions

2. Average

10. Documentation (Rate the paper organize as per the IEEE format, clarity of text, figures and accurarcy of references)

3. Above Average

11. Comments to Authors

1. The evaluation of the agentic layer is relatively limited. While classification accuracy and proposed agentic metrics are reported, there is no statistical significance analysis or confidence intervals to demonstrate the robustness of the observed improvements.
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12. Recommendation
2. Accept

Reviewer #4

Questions

1. Technical content and scientific rigour (Rate the technical .content of paper)
3. above average

2. 2. Novelty and Contributions
3. Above Average

3. 3. Documentation (Rate the paper organize as per the IEEE format, clarity of text, figures and accurarcy of references)
2. Average

4. 4. Comments to Authors
The novelty of the proposed multi-agent framework should be clearly distinguished from existing agentic AI and plant disease diagnosis approaches.
The roles, interactions, and decision boundaries of the four agents should be described more precisely.
The criteria used by the Confidence and Image-Quality Agent to trigger human intervention should be quantitatively justified.
The Retrieval Agent should be evaluated for the accuracy, reliability, and quality of the evidence retrieved.
The reasoning process used to generate treatment recommendations should be validated against expert-agreed recommendations.
The claim of “autonomous” diagnosis should be moderated given that low-confidence and ambiguous cases are explicitly deferred to human experts.
The use of PlantVillage alone may not adequately demonstrate robustness under real-world field conditions with variations in lighting, background, disease severity, and image quality.
The reported 99.1% accuracy should be supported with additional evaluation metrics and statistical significance analysis.
A comprehensive ablation study is needed to quantify the individual contribution of the vision, retrieval, reasoning, and orchestration components.
The proposed agentic metrics—correct-action rate, harmful-action rate, and human-intervention rate—should be formally defined and evaluated against meaningful baselines.

The harmful-action rate requires particularly careful evaluation because incorrect treatment recommendations may have significant agricultural and economic consequences.
The federated learning experiments using FedAvg, FedOpt, and FedMA should be better connected to the proposed multi-agent framework and their practical motivation should be clarified.
The system should be evaluated on external or cross-dataset data to establish its generalization beyond the PlantVillage benchmark.
The computational overhead, latency, and resource requirements introduced by the retrieval and multi-agent reasoning components should be reported.
The paper should discuss potential failure cases, hallucinated recommendations, outdated retrieved evidence, and safeguards against unsafe treatment advice.

5. 5. Recommendation

2. Accept

8. Technical content and scientific rigour (Rate the technical content of paper)

3. above average

9. Novelty and Contributions

3. Above Average

10. Documentation (Rate the paper organize as per the IEEE format, clarity of text, figures and accurarcy of references)

4. High

11. Comments to Authors

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